

Research Completion Report

Project: BRE Concrete Mix Design Engineering Knowledge Base **Date:** 2026-09-18 **Phase:** Phase 1 - Research & Engineering Documentation

1. What Was Researched

The complete BRE/DoE concrete mix-design method as documented in BRE BR 331, Design of Normal Concrete Mixes, Second Edition, 1997. Additionally, the relationship between this historical mix-design method and current UK concrete specification standards (BS 8500, BS EN 206) was researched and documented.

2. Sources Found

ID	Source	Tier	Status
SRC-BRE-001	BRE BR 331, 2nd Ed, 1997	1	VERIFIED (PDF obtained)
SRC-BSI-001	BS 8500-1:2023	1	VERIFIED
SRC-BSI-002	BS 8500-2:2023	1	VERIFIED
SRC-BSEN-001	BS EN 206	1	VERIFIED
SRC-BSEN-002	BS EN 197-1	1	VERIFIED
SRC-BSEN-003	BS EN 12620	1	VERIFIED
SRC-CS-001	Concrete Society	2	VERIFIED
SRC-CC-001	Concrete Centre	2	VERIFIED
SRC-TRID-001	TRID/TRB	3	VERIFIED
SRC-UNI-001	EMU Lecture Notes	3	SECONDARY

Total: 10 sources catalogued

3. Sources Verified

- BRE BR 331: PDF downloaded from official BRE Group website [X]
- ISBN 1-86081-172-8 confirmed via TRID [X]
- BS 8500:2023 current editions confirmed via BSI/Concrete Centre [X]
- All Tier 1 sources confirmed for existence and relevance [X]

4. BRE Method Sections Documented

File	Content	Status
00_OVERVIEW.md	Method overview, 5-stage workflow	Complete
01_SCOPE.md	Assumptions (A1-A7), Limitations (L1-L7)	Complete
02INPUTREQUIREMENTS.md	19 engineering inputs + validation rules	Complete
03TARGETMEAN_STRENGTH.md	EQ-BRE-001/002, k factors, worked example	Complete
04WATERCEMENT_RATIO.md	Table 2, Figure 4 (partial digitization)	Complete
05WATERCONTENT.md	Table 3, workability classification	Complete
06CEMENTCONTENT.md	EQ-BRE-003, min/max checks	Complete
07TOTALAGGREGATE.md	EQ-BRE-004, Figure 5 (partial digitization)	Complete

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File	Content	Status
08FINEAGGREGATE.md	EQ-BRE-005, Figure 6 (partial digitization)	Complete
09COARSEAGGREGATE.md	EQ-BRE-006	Complete
10_ADJUSTMENTS.md	Trial mix, durability, combined aggregate	Complete
11FIELDCORRECTIONS.md	EQ-BRE-007/008/009	Complete
12FINALMIX.md	Output format, checks, report template	Complete
13SPECIALCASES.md	Air-entrained, PFA, GGBS	Complete
99_TRACEABILITY.md	All items mapped to sources	Complete

5. Equations Extracted

ID	Name	Status
EQ-BRE-001	Target mean strength	CROSS-VERIFIED
EQ-BRE-002	Margin	CROSS-VERIFIED
EQ-BRE-003	Cement content	VERIFIED
EQ-BRE-004	Total aggregate	VERIFIED
EQ-BRE-005	Fine aggregate	VERIFIED
EQ-BRE-006	Coarse aggregate	VERIFIED
EQ-BRE-007	Batch water	CROSS-VERIFIED
EQ-BRE-008	Batch fine aggregate	CROSS-VERIFIED
EQ-BRE-009	Batch coarse aggregate	CROSS-VERIFIED

Total: 9 equations extracted and documented

6. Tables Extracted

ID	Name	Status
TBL-BRE-001	Workability classification	CROSS-VERIFIED
TBL-BRE-002	Compressive strength at w/c 0.5	CROSS-VERIFIED
TBL-BRE-003	Free water content	CROSS-VERIFIED

Total: 3 tables extracted; 2 with JSON machine-readable data

7. Graphs Identified

ID	Figure	Digitization Status
GRAPH-BRE-001	Figure 4: Strength vs w/c ratio	PARTIAL
GRAPH-BRE-002	Figure 5: Wet density	PARTIAL
GRAPH-BRE-003	Figure 6: Fine aggregate proportion	PARTIAL

Total: 3 graphs identified; all require complete digitization from physical source

8. Data Digitized

- Table 2 values: 4 rows x 4 age columns = 16 data points [X]
- Table 3 values: 6 rows x 4 slump columns = 24 data points [X]
- k factor values: 4 values [X]
- Figure 4: ~11 data points for one cement/aggregate combination
- Figure 5: ~28 data points (estimated)
- Figure 6: ~72 data points (2 of 12 sub-tables)

9. Worked Examples Found

- Constructed 2 golden test cases from the BRE method steps
- CASE-001: C30, 42.5 cement, uncrushed 20mm
- CASE-002: C40, 42.5 cement, crushed 20mm, with durability constraints

10. Validation Cases Created

Type	Count	Status
Golden test cases	2	Created
Equation tests	6	Planned
Component tests	7	Planned
Boundary tests	10	Planned
Invalid input tests	8	Planned
Independent verification	3	Planned

11. Current Standards Reviewed

- BS 8500-1:2023 and BS 8500-2:2023 - confirmed as current UK standards
- BS EN 206 - European framework standard identified
- BS EN 197-1 - cement classification confirmed
- BS EN 12620 - aggregate standard identified
- Relationship between BRE and current standards documented
- One conflict identified and resolved (CONFLICT-001: cement classification)

12. Conflicts Identified

ID	Topic	Status
CONFLICT-001	Cement classification (old vs current)	RESOLVED - BR 331 2nd ed already uses 42.5/52.5

13. Unverified Information

Item	Reason	Priority
Standard deviation guidance table	Not directly extracted from PDF	MEDIUM
Coarse aggregate fraction proportions	Not found in available sources	LOW
Mixed aggregate water formula exact wording	SECONDARY source only	LOW

14. Engineering Review Items

Item	Description	Priority
Figure 4 complete digitization	Require physical copy for accurate curve data	HIGH
Figure 5 complete digitization	Require physical copy for chart data	HIGH

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Item	Description	Priority
Figure 6 complete digitization	Only 20mm/2 slump ranges done; need all 12 sub-tables	HIGH
Standard deviation defaults	Verify exact table from BR 331	MEDIUM
Rounding rules	Many marked UNSPECIFIED	LOW

15. Missing Information

Item	Impact	Mitigation
Complete Figure 4 curve data	Cannot accurately determine w/c ratio	Use approximation formula + flag
Complete Figure 5 chart data	Approximate wet density only	Use linear approximation + flag
Complete Figure 6 for 10mm and 40mm	Cannot proportion for these sizes	Partial digitization available for 20mm
Worked examples from BR 331	Golden tests are constructed, not original	Cross-verified intermediate values

16. Recommended Next Phase

Immediate (before coding):

1. Obtain physical copy of BR 331 - for complete digitization of Figures 4, 5, 6
2. Complete Figure 6 digitization for all 3 aggregate sizes x 4 slump ranges = 12 sub-tables
3. Verify standard deviation guidance table from BR 331

Phase 2 (calculation engine):

1. Implement calculation engine in Python (no UI)
2. Implement all equations (EQ-BRE-001 to EQ-BRE-009)
3. Implement all table lookups
4. Implement Figure 4/5/6 interpolation
5. Run golden tests
6. Run boundary tests

17. Overall Research Completeness

Area	Status	Confidence	Notes
BRE method	Complete	High	All 5 stages documented with equations
Equations	Complete	High	9 equations, all verified
Tables	Complete	High	3 tables, cross-verified, JSON created
Graphs	Partial	Medium	3 identified; partial digitization
Standards	Complete	High	Relationships documented
Validation	Structured	Medium	2 golden cases; 34 tests planned
Traceability	Complete	High	Full matrix created
Software spec	Complete	High	Architecture, engine, UI, DB, reports

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Area	Status	Confidence	Notes
AI governance	Complete	High	Rules, workflow, guardrails, templates

Files Created Summary

Directory	Files Created
Root	2 (AGENTS.md, README.md)
00_PROJECT/	5 (scope, roadmap, requirements, terminology, decisions, changelog)
01_SOURCES/	2 (source register, BR331 review)
02BREMETHOD/	15 (overview through traceability)
03_STANDARDS/	1 (standard relationships)
04ENGINEERINGDATA/	4 (data dictionary, constants, 2 JSON tables)
05_CALCULATIONS/	6 (pipeline, equations, interpolation, rounding, units, errors)
06_VALIDATION/	3 (validation plan, 2 golden case JSON files)
07SOFTWARESPEC/	6 (architecture, engine, UI, DB, reports, traceability)
08AIAGENT/	7 (rules, workflow, template, guardrails, checklist, protocol, audit)
Total	~51 files

Report generated: 2026-09-18 Phase 1 Status: COMPLETE (with engineering review items noted)